Rishit Mittal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.32
Shri Krishna Public School	Class XII (CBSE)	2024	92.40%
Indore Public School	Class X (CBSE)	2022	95.00%

SCHOLASTIC ACHIEVEMENTS

INMO 2024: Selected for the national stage by clearing the RMO cutoff from a pool of ~80,000 IOQM candidates.

INChO 2024: Selected for the national stage, ranking among the top 531 out of 44,363 candidates nationwide.

NSEC 2023: Awarded the State Merit Certificate for securing a rank in the top 1% of 1857 candidates of the state.

NSEP 2024: Awarded the State Merit Certificate for securing a rank in the top 1% of 2074 candidates of the state.

PROJECTS

Sentiment Classification NLP Framework | Python, PyTorch

- **Built an NLP text classification pipeline** to benchmark 16 different RNN and LSTM architectures on a 50,000-sample dataset, reaching a peak validation accuracy of 91.4% and a 0.88 F1-score.
- **Implemented custom Bahdanau and Luong attention layers**--the foundational architecture behind modern LLMs--to extract interpretable heatmaps and improve how the model processes text context.
- **Streamlined model inference and training** by developing an end-to-end PyTorch data pipeline to efficiently handle custom tokenization, dynamic batch padding, and out-of-vocabulary words.
- **Fine-tuned model convergence** and prevented overfitting on text embeddings by applying AdamW optimization, learning rate schedulers, and dropout.

Project Vani: Offline Hindi Voice Assistant Pipeline | Python, PyTorch, ONNX

- Developed a local speech-to-text and intent-parsing pipeline using a **Wav2Vec2** transformer model for Hindi ASR and **eSpeak-NG** for automated feedback.
- Simulated a 3x inference speedup under ARM-based constraints by converting the PyTorch architecture into a **quantized INT8 ONNX** format.
- Integrated **Voice Activity Detection (VAD)** into the input stream to filter idle noise and optimize CPU cycles by triggering inference only during active speech.

Statistical Data Analysis (Part of Applied Statistics coursework) |(Link to reports)

- Performed statistical inference using sampling, point/interval estimation, confidence intervals, and hypothesis testing.
- Applied t-tests, Chi-square tests, MLE, MOM, and correlation analysis using Python on real-world datasets.
- Visualized and modeled data with Matplotlib, validating the Central Limit Theorem and fitting Gamma distributions.

xv6 OS Kernel Architecture & Extensions | C,C++, RISC-V Assembly

- **Eliminated thread starvation and prioritized interactive tasks** by replacing the default round-robin scheduler with a 4-level, syscall-aware Multi-Level Feedback Queue (MLFQ).
- **Achieved O(1) physical memory lookups** and reduced page fault latency by approximately 40% by engineering a demand paging system utilizing RISC-V PTE_A bits and a static Frame Table.
- **Guaranteed zero-deadlock execution** across multi-core CPUs by implementing custom system calls mapped to a strict hierarchical spin-lock protocol.
- **Enabled fault-tolerant data recovery** by building a 3-Zone logical storage layer supporting RAID 0/1/5 arrays and SSTF disk scheduling.

RISC-V Pipeline and Cache Simulator | C++

- Resolved critical data race conditions and achieved cycle-accurate instruction throughput by engineering double-buffered pipeline registers for a **custom 5-stage RISC-V CPU simulator.**
- Minimized memory access latency and achieved a **92% cache hit rate by integrating a multi-level L1/L2** associative cache hierarchy configured with a write-back memory policy.
- Reduced control hazard penalties and improved branch prediction accuracy to **88% by integrating a Branch Target Buffer (BTB)** coupled with a 2-bit saturating counter and also experimented with using the **G-share.**
- Maintained uninterrupted pipeline flow during data conflicts by designing a hardware **Hazard Detection Unit** equipped with custom ALU operand forwarding and stall-cycle injection logic.

SKILLS

Languages: C++, C, Python, Assembly(RISC-V), SQL, HTML, CSS,Java

Machine Learning: Scikit-Learn, Tensorflow, Numpy, Pandas, PyTorch

DEV Tools/Systems: Linux,QEMU,Git, GitHub, Docker

RELEVANT COURSES

Computer Science: Operating Systems, Software Development Fundamentals, Computer Architecture, Data Structure, Algorithms, Database Management Systems, Theory of Computation, Compilers-I,Discrete Mathematics, Artificial Intelligence

Mathematics: Calculus -I, II, Probability and Random variables, Linear Algebra, Transform techniques, Applied Statistics

EXTRACURRICULAR

- Core Member - RDM, Dramatics and Speaking Arts Club, IITH
- Participated in INTER IIT Culturals 8.0, held in IIT Kanpur